

Cross-Network Effects Between Perceived Slices and Self-Reported Network

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1 Overview

We implemented cross-network effects between perceived slices $Y^{(p)}$ and the self-reported network $Y^{(\text{self})}$ by reusing RSiena’s existing `crprod` infrastructure (the `nonSymmetricNonSymmetricObjective` effect group), avoiding any C++ changes.

Notation. The CSS tensor is x_{pjk} where p = perceiver, j = sender, k = receiver. The perceived slice for perceiver p is the $n \times n$ one-mode network $Y_{jk}^{(p)} = x_{pjk}$, and the self-reported network is $Y_{jk}^{(\text{self})} = x_{jjk}$. In the standard RSiena two-network notation (manual Section 12.1.3), the dependent network is $X = Y^{(p)}$ and the explanatory network is $W = Y^{(\text{self})}$. The focal actor making a minisstep in $Y^{(p)}$ is sender j ($j \neq p$, since row p is structurally fixed).

2 Dyadic Cross-Network Effects

2.1 Entrainment / `crprod` (`crprod`)

RSiena formula (manual effect 1):

$$s_j^{\text{net}}(x) = \sum_k x_{jk} w_{jk}.$$

Three-way form:

$$s_j^{(p)}(x) = \sum_k x_{pjk} \cdot x_{jjk}.$$

Interpretation. Perceptual accuracy: how much perceiver p ’s view of j ’s outgoing ties matches j ’s own self-reported ties. A positive parameter means the perceived network tends to align with the self-report.

2.2 Reciprocal entrainment (`crprodRecip`)

RSiena formula (manual effect 2):

$$s_j^{\text{net}}(x) = \sum_k x_{jk} w_{kj}.$$

Three-way form:

$$s_j^{(p)}(x) = \sum_k x_{pjk} \cdot x_{kkj}.$$

Interpretation. Generalised exchange: the existence of k 's self-reported tie to j (i.e. k says "I have a tie to j ") promotes perceiver p perceiving that j has a tie to k . A positive parameter indicates that being named by k makes it more likely that j is perceived as sending a tie to k .

2.3 Mutual entrainment (crprodMutual)

RSiena formula (manual effect 3):

$$s_j^{\text{net}}(x) = \sum_k x_{jk} w_{jk} w_{kj}.$$

Three-way form:

$$s_j^{(p)}(x) = \sum_k x_{pjk} \cdot x_{jjk} \cdot x_{kkj}.$$

Interpretation. Accuracy conditional on mutual self-report: the perceived tie $j \rightarrow k$ aligns with reality only when both j and k self-report ties to each other. A positive parameter suggests that mutual self-reported ties are especially visible to perceivers.

2.4 Entrainment $\times$ in-degree activity (crprodInActIntn)

RSiena formula (manual effect 13):

$$s_j^{\text{net}}(x) = \sum_k x_{jk} w_{jk} (\sqrt{w_{+j}} - \sqrt{\bar{w}}), \quad p = 2.$$

Three-way form:

$$s_j^{(p)}(x) = \sum_k x_{pjk} \cdot x_{jjk} \cdot \left(\sqrt{\sum_\ell x_{\ell\ell j}} - \sqrt{\bar{w}} \right),$$

where $\sum_\ell x_{\ell\ell j}$ is the self-reported in-degree of j (how many others report a tie to j), and $\bar{w}$ is the mean degree of $Y^{(\text{self})}$.

Interpretation. Moderation of perceptual accuracy by self-reported popularity: the alignment between perceived and self-reported ties is stronger for actors who are popular (high in-degree) in the self-reported network.

3 Degree-Related Cross-Network Effects

The following effects involve degree statistics of $W = Y^{(\text{self})}$. In all cases, W -degrees are centered by the average degree $\bar{w}$ of $Y^{(\text{self})}$.

3.1 W in-degree on X -popularity (inPopIntn)

$$s_j^{(p)}(x) = \sum_k x_{pjk} (\sqrt{w_{+k}} - \sqrt{\bar{w}}), \quad w_{+k} = \sum_\ell x_{\ell\ell k}.$$

Actors who are popular in the self-reported network attract more perceived incoming ties.

3.2 W in-degree on X -activity (inActIntn)

$$s_j^{(p)}(x) = x_{pj+} (\sqrt{w_{+j}} - \sqrt{\bar{w}}), \quad w_{+j} = \sum_\ell x_{\ell\ell j}.$$

Actors who are popular in the self-reported network are perceived as more active (higher perceived out-degree).

3.3 W out-degree on X -popularity (outPopIntn)

$$s_j^{(p)}(x) = \sum_k x_{pjk} (\sqrt{w_{k+}} - \sqrt{\bar{w}}), \quad w_{k+} = \sum_\ell x_{kk\ell}.$$

Perceived ties tend toward actors who are active (high out-degree) in the self-reported network.

3.4 W out-degree on X -activity (outActIntn)

$$s_j^{(p)}(x) = x_{pj+} (\sqrt{w_{j+}} - \sqrt{\bar{w}}), \quad w_{j+} = \sum_\ell x_{jj\ell}.$$

Actors who are active in the self-reported network are perceived as more active.

4 Mixed Triadic Effects

These effects capture transitive-like configurations across the perceived and self-reported networks. Below, $X = Y^{(p)}$ and $W = Y^{(\text{self})}$.

4.1 From W agreement (from)

$$s_j^{(p)}(x) = \sum_{h \neq k} x_{pjk} w_{jh} w_{kh} = \sum_{h \neq k} x_{pjk} x_{jjh} x_{khh}.$$

Structural equivalence in outgoing self-reports: perceived tie $j \rightarrow k$ is promoted when j and k self-report ties to the same third actors h (i.e. j and k agree on who to nominate).

4.2 W to agreement (to)

$$s_j^{(p)}(x) = \sum_{h \neq k} x_{pjk} w_{hj} x_{phk} = \sum_{h \neq k} x_{pjk} x_{hhj} x_{phk}.$$

Mixed W - X two-path closure: if there is a self-reported tie $h \rightarrow j$ and a perceived tie $h \rightarrow k$ by perceiver p , this promotes the perceived tie $j \rightarrow k$.

4.3 Closure of W (closure)

$$s_j^{(p)}(x) = \sum_{h \neq k} x_{pjk} w_{jh} w_{hk} = \sum_{h \neq k} x_{pjk} x_{jjh} x_{hkh}.$$

Closure of W - W two-paths: if j self-reports a tie to h and h self-reports a tie to k , this promotes perceiver p perceiving a tie $j \rightarrow k$. “Self-reported two-paths tend to become perceived ties.”

4.4 XWX closure (cl.XWX)

$$s_j^{(p)}(x) = \sum_{h \neq k} x_{pjk} x_{pjh} w_{hk} = \sum_{h \neq k} x_{pjk} x_{pjh} x_{hkh}.$$

If perceiver p sees $j \rightarrow h$ and h self-reports a tie to k , this promotes perceiver p also perceiving $j \rightarrow k$.

5 Implementation Notes

1. All effects are generated automatically by calling `createEffects("nonSymmetricNonSymmetricObjective", selfName, name = sliceName)` inside `threeWayNet()` in `effects.r`.
2. The shared parameter framework applies: $Y[\text{shared}]$ is the canonical parameter holder (`sharedDup = FALSE`), and $Y[2], \dots, Y[K]$ are hidden duplicates (`sharedDup = TRUE`).
3. Usage example:

```
eff <- includeEffects(eff, crprod,
                     interaction1 = "Y[self]",
                     name = "Y[shared]")
```

4. No C++ modifications were needed. The C++ side resolves `pState->pNetwork("Y[self]")` by name string, and `initializeFRAN.r` already registers $Y^{(\text{self})}$ as a separate one-mode network.

6 Bug Fix: Structural Zeros in Perceived Slices

We identified a bias in the parameter recovery simulation (true $\beta = 0.6$, estimated $\hat{\beta} \approx 0.4$). The cause was that the structural-10 marking of perceiver p 's own row in slice $Y^{(p)}$ was applied only in `effects.r` (for effect-table generation) but *not* in `initializeFRAN.r` (for the actual C++ data).

Without structural zeros on row p of $Y^{(p)}$:

- Actor p could make ministeps in $Y^{(p)}$, modifying x_{ppk} .
- The same cells x_{ppk} also live in $Y^{(\text{self})}$, causing double-counting.
- The `crprod` statistic was inflated by the $j = p$ self-correlation term $\sum_k (x_{ppk})^2$, leading to downward bias in $\hat{\beta}_{\text{crprod}}$.

Fix: In `initializeFRAN.r`, `.expandThreewayToOneModes()`, after extracting `arr3 <- depvar[k,,]`, we now set `arr3[k, , ] <- 10` (structural zero) for all time points. The distance computation was also updated to exclude row k .